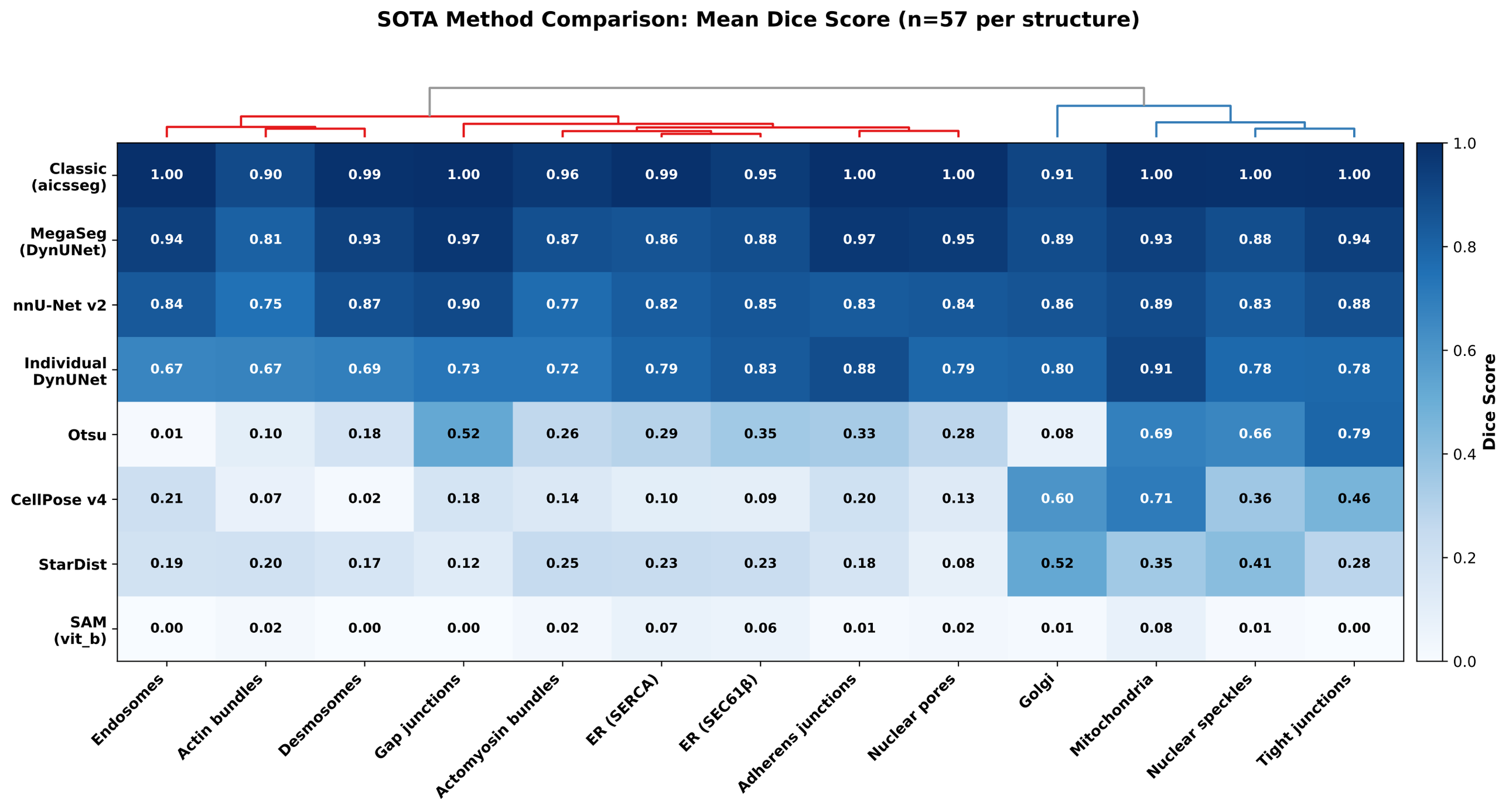

SOTA Method Comparison: Mean Dice Score (n=57 per structure)
Classic (aicsseg)
MegaSeg (DynUNet)
nnU-Net v2
Individual DynUNet
Otsu
CellPose v4
StarDist
SAM (vit_b)
1.00 0.90 0.99 1.00 0.96 0.99 0.95 1.00 1.00 0.91 1.00 1.00 1.00
0.94 0.81 0.93 0.97 0.87 0.86 0.88 0.97 0.95 0.89 0.93 0.88 0.94
0.84 0.75 0.87 0.90 0.77 0.82 0.85 0.83 0.84 0.86 0.89 0.83 0.88
0.67 0.67 0.69 0.73 0.72 0.79 0.83 0.88 0.79 0.80 0.91 0.78 0.78
0.01 0.10 0.18 0.52 0.26 0.29 0.35 0.33 0.28 0.08 0.69 0.66 0.79
0.21 0.07 0.02 0.18 0.14 0.10 0.09 0.20 0.13 0.60 0.71 0.36 0.46
0.19 0.20 0.17 0.12 0.25 0.23 0.23 0.18 0.08 0.52 0.35 0.41 0.28
0.00 0.02 0.00 0.00 0.02 0.07 0.06 0.01 0.02 0.01 0.08 0.01 0.00
Endosomes
Actin bundles
Desmosomes
Gap junctions
Actomyosin bundles
ER (SERCA)
ER (SEC61β)
Adherens junctions
Nuclear pores
Golgi
Mitochondria
Nuclear speckles
Tight junctions
Dice Score
1.0
0.8
0.6
0.4
0.2
0.0